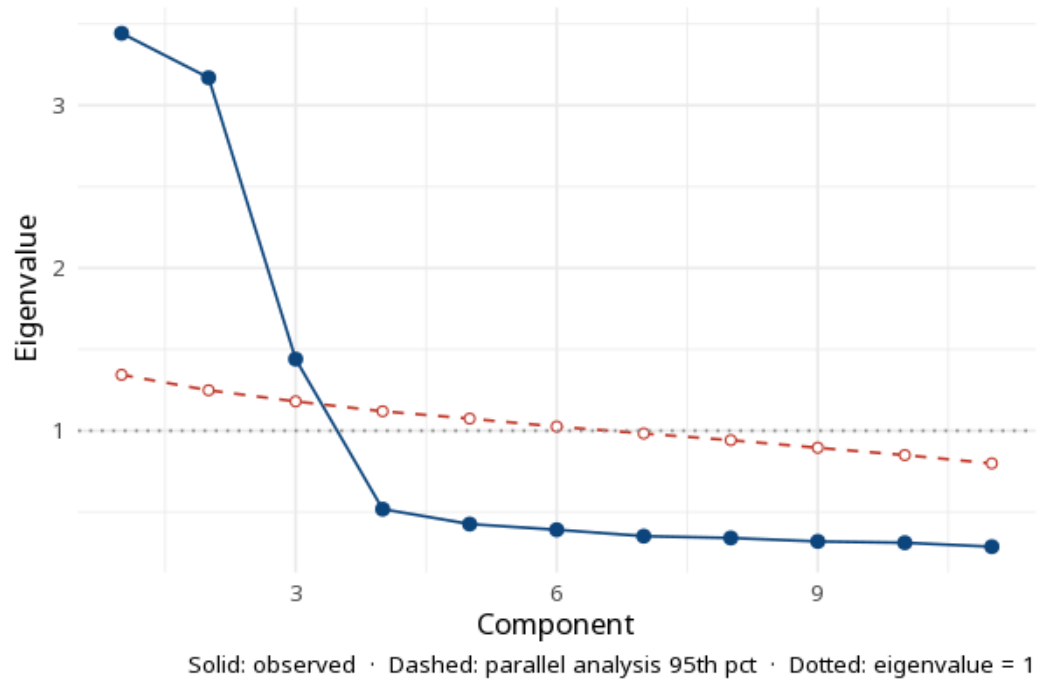


1 Principal component analysis (PCA)

Component	Eigenvalue	% variance	Cumulative %
PC1	3.443	31.3	31.3
PC2	3.170	28.8	60.1
PC3	1.440	13.1	73.2

Variable	PC1	PC2	PC3
esg1	-.62	-.51	
esg2	-.56	-.52	.33
esg3	-.58	-.52	.33
esg4	-.56	-.51	
norm1	-.39	.75	
norm2	-.39	.71	
norm3	-.41	.72	
norm4	-.41	.72	
intent1	-.71		-.53
intent2	-.66		-.56
intent3	-.72		-.48



PCA compresses correlated variables into a few components. The variance explained (Table 1) shows how much information each component retains; the component loadings (Table 2) show how strongly each variable contributes.

Loadings are correlation-scaled ($\text{eigenvector} \times \sqrt{\text{eigenvalue}}$) so they share the unit of a correlation coefficient. PCA is data reduction — components are weighted composites of the observed variables, not reflective latent factors; do not interpret them as constructs and do not report communalities (which belong to factor analysis, not PCA). For a latent-construct model use EFA (psych::fa) or CFA (lavaan). The scree plot overlays the parallel-analysis 95th-percentile threshold — retain components where the observed eigenvalue exceeds that line; the Kaiser eigenvalue > 1 rule is shown for reference but tends to over-extract (Zwick & Velicer, 1986). Cumulative variance $\geq 70\%$ is a common but explicitly subjective target (Jolliffe & Cadima, 2016).

PCA (correlation matrix; parallel analysis) retained 3 components explaining 73.2% of total variance.

Statistical basis: Principal component analysis as a data-reduction method (not a latent-variable model) (Jolliffe, I. T., Cadima, J. (2016). "Principal component analysis: a review and recent developments." *Philosophical Transactions of the Royal Society A*, 374, 20150202.); PCA is not EFA - components are composites, not reflective factors (Frick, S., Killisch, R., BiSSantz, N., Wetzel, E. (2025). "PCA Is Not EFA." *European Journal of Psychological Assessment*, 41(6), 425-426.); Kaiser eigenvalue > 1 rule tends to over-extract (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." *Psychological Bulletin*, 99(3), 432-442.).